Pole-chart leading factors and actual anchor degree budget

14 November 2026; source `43b457291f5b99ae1c23c4057b34ad07bdb07b41`

Component results. The same completed orientation family has explicit polynomial leading factors at its retained negative-pole charts, with 2^u labels. Its actual diagonal degree also bounds the last retained anchor multiplicity by a Fibonacci factor. The pinned shared gate audits 59 declarations with zero extra trusted axioms. Import-inclusive fresh replay is running at this report snapshot; no completed exit is claimed. All records are unsigned [pcc:unverified].

Actual chart, every coefficient. Let $L = 2^a$, $i < u \le a$, $t = -2^i$, $\alpha = 2^{a+1}$, and

$$M_{xy} = \{j < u : \mathrm{bit}_j(x) = \mathrm{bit}_j(y)\}, \quad G_i = m_i - \sum_{j<u} (j-i)_+ m_j.$$

Use the original actual completed polynomial $W_{xy}(T)$ on actual ports $x, y < 2^a$, and equal-bit-i crossings. Let $P_{xy}(Z) = W_{xy}(t + \alpha Z)$, $J_{xy} = [Z]P_{xy}$, $c = J_{00} \neq 0$, and $w = v_2(c) = a + 1 - i + G_i$. No favorable scalar is supplied. Multiplicities are natural numbers; the matrix leading-factor statements additionally require $m_j > 0$ for every $j < u$. The finite rest product and actual anchor slopes construct a polynomial U_{xy} such that

$$P_{xy} = J_{xy} Z U_{xy}, \qquad U_{xy}(0) = 1, \qquad v_2([Z^n]U_{xy}) \ge n$$

with zero coefficients admitted. The full chart has content at least w, so J_{xy}/c is integral. Therefore

$$v_2([Z^{n+1}](P_{xy}/c)) \ge n, \qquad P_{xy}/c \equiv (J_{xy}/c)Z \pmod 2$$

coefficientwise. This is a universal symbolic statement at these specified negative poles, not a sample of coefficients [pcc:unverified].

Constructed matrix lift. Zero entries outside the crossings. The parent's actual first-jet factors A, B satisfy $R - AB \equiv 0 \pmod 2$, $R_{xy} = J_{xy}/c$. Set $\widetilde{A} = C(A)$ and $\widetilde{B} = C(B)Z$, with the same 2^u labels. Then

$$P/c - \widetilde{A}\widetilde{B} \equiv 0 \pmod 2.$$

Both factors are integral. Scaling the left by $C(c)$ gives row content w and actual residual content $w + 1$; the scaled left is integral if $w \ge 0$. This covers the lowest precision layer of the whole polynomial chart, not arbitrary precision or all active source nodes [pcc:unverified].

A necessary budget for the actual polynomial

This page uses $n = u + 1$ retained levels, with terminal level u; this is an index convention, not an extra retained level in the chart theorem. Construct the actual list $(m_0, \dots, m_u)$ from the finite range. The Lean bridge derives the abstract recurrence from the actual inequalities $G_i \geq 0$ for every $i < u + 1$. A caller does not supply a feasible list. For a list $\ell = (m_0, \dots, m_{n-1})$, its weighted suffix penalty is

$$\operatorname{penalty}(\ell) = \sum_{j<n} (j+1) m_j.$$

Prepending a multiplicity $m \geq P$ to an existing suffix of mass S and penalty P gives

$$S' = m + S \geq P + S, \qquad P' = m + P + S \geq 2P + S.$$

Starting from the actual terminal multiplicity $r = m_u$, the initial suffix has $S = P = r$. Fibonacci induction gives

$$F_{2u+1} m_u \leq \sum_{j<u+1} m_j, \qquad F_{2u+2} m_u \leq \operatorname{penalty}(m_0, \dots, m_u).$$

These are necessary inequalities for this explicit schedule, not a lower bound for arbitrary completion polynomials [pcc:unverified].

Exact degree, not reversal of an upper bound. For a finite source set $S \subset \mathbb{Q}$ with $0 \in S$, the actual normalizer is nonzero and the diagonal base has degree $|S| - 1$. Every selected factor is nonzero of degree one. At $x = y$ the actual mask selects all retained anchors. Degrees add over $\mathbb{Q}$, giving

$$\deg W_{xx} = |S| - 1 + \sum_{j<u+1} m_j.$$

For the actual signed grid $|S| = 2L$, an actual diagonal cap

$$\deg W_{xx} \leq 2L - 1 + KL \quad \Longrightarrow \quad F_{2u+1} m_u \leq KL.$$

The previous upper-degree estimate alone would not justify this direction. Nonzero selected factors and the actual normalizer are derived in the proof [pcc:unverified].

Proof map. Full statements, source bindings and all 54 new unsigned claim ids are in `../evidence/claim-pins.json`. Central declarations:

`actual_completed_chart_unit`	PCC-20261114-4e9529
`actual_normalized_coefficient_weight`	PCC-20261114-0be10e
`actual_chart_leading_factorization`	PCC-20261114-141d98
`actual_weighted_chart_leading_factorization`	PCC-20261114-dc6a99
`actual_schedule_feasible`	PCC-20261114-d08950
`completed_diagonal_degree`	PCC-20261114-b4a0e7
`actual_terminal_polynomial_budget`	PCC-20261114-2d4b00

All are unsigned/default not covered; the full Lean statement governs each claim's hypotheses [pcc:unverified].

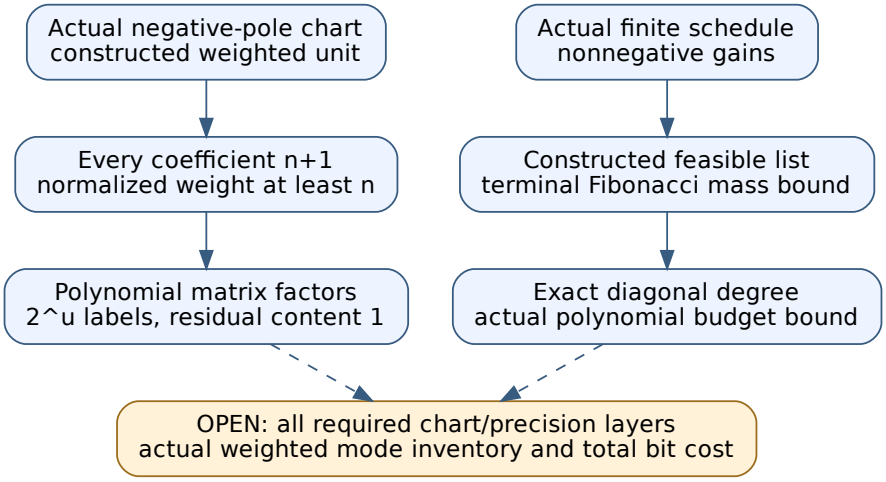
Mode cost: the comparison that must still be made

The pinned primary proof D.4.3–D.4.7 has $D_i = L/2^i$, at most $3M/L$ packets, precision $p_t = O(MD_i/u)$, and at most $6M/D_i$ modes per weight. Its weighted tuple bound is $\exp(O(\sqrt{Mp_t/D_i}))$, yielding $\exp(O(M/\sqrt{u}))$. These are the reviewed primary proof's formulas, not a new measured or independently formalized runtime result [pcc:unverified].

For the separate actual first-jet moment construction, every residue class r mod 2^u and every retained moment b is charged. The corresponding *declared* per-weight inventories and bounds are:

	Original primary profile	Residue moment profile
labels per packet per moment	2^{i+1}	2^u
public row weight	$b+1$	$w+b$
packet-count upper budget	$6M/D_i$	$(3M/L)2^u$

The ratio of those declared upper budgets is 2^{u-i-1}. It is not an observed count ratio or a lower bound for the number of independent nonzero modes. Higher coefficient valuations, zero coefficients or a different basis could change the relevant inventory. A smaller symbolic factor count than row refinement does not settle the original comparison [pcc:unverified].



Next discriminating obligation. Construct and charge the actual weighted mode inventory, including coefficient construction and target/source locality. Combine it with the exact degree cap, required precision and recovery loss. The negative-pole lowest-layer statement cannot be substituted for all required charts or higher precision. A full prime-field ECDLP proof also needs the actual relation map, public finder, field/scalar extraction, retry bounds and complete expected bit cost.

Evidence, scope and reproducibility

Checks. The source contains 57 authored statements: 33 pole-chart and 24 budget statements. Three are copied historical budget lemmas; 54 are new source-bound declarations. The shared gate audits 59 declarations, including these generated declarations:

```
AnchorBudget.Feasible.brecOn
LocalCharts.localBase.eq_1
```

All three new tracked modules were built and audited. Local build and gate exit 0. Import-inclusive fresh replay is running at this snapshot:

```
leanchecker -fresh -verbose OrientationCompletion
```

No completed fresh exit is claimed [pcc:unverified].

Pinned dependencies. Lean 4.35.0, with:

CryptoLean `148f5dc389b341401e06d76abe7783c4ccf42f2e`
Mathlib `cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`

Exact source, manifest, gate, audit and trust hashes are in `../evidence/source-freeze.json`. Gate data is PCC-20261114-67cb41. Local draft records 623aab, 92e0dc, b83c13, a4c256 and 73e030 preserve intermediate status. The axioms shown are only `propext`, `Classical.choice` and `Quot.sound` [pcc:unverified].

Failures retained. Chart 006 rewrite/visibility and 008 helper normalization failures; budget 001 path failure, 002 recurrence/normalization, 003 zero-add, 004 unused-tactic warnings and 007 factor-unfolding failure. All raw source/log/exit versions remain under the moment evidence directory. Rejected drafts' sorryAx diagnostics are failures, not accepted proof inputs. No warnings were suppressed; frozen parent source and report checkpoints were not changed. There was no native cryptographic, timing, scaling or paid/cloud experiment in this round.

Sources checked. KB unavailable: not loaded, wrapper has no executable; the journal records bounded GitHub fallback queries. Read the original AnchorBudget, actual OrientationAnchor/Leading/Jet and LocalCharts/Integral sources, pinned Mathlib list/degree/coefficient APIs, and the primary D.4.3–D.4.7 with source-triage, contraction-cost and contraction-precision reviews under

```
research/prime_one_shot_20261029/continuation/tensor-review/
```

Primary PDF SHA256, split for legibility: `5b1612dc2eebb2cad5b4cbe62d1fed2982`
`f01ab1908c26f70010a651487e8123`. Standard inverse moments, Fibonacci induction and polynomial degree laws are reused; there is no priority or fastest-method assertion.

Graph/evidence policy. This is a symbolic proof-component report, so the graphic is a dependency diagram, not an empirical plot. The literal parent ring-scaling certificate IDC1h57849461da3c8854 remains limited to its fixed 32+16 points and is not a certificate of these parameterized valuations, Fibonacci bounds or runtime. No new measurement supports changes to the IC scoreboard/timeline/leaderboard or historical parent figures; these stay unchanged. Editable DOT/SVG/PDF and this TeX/PDF are kept together.

Integration. PR4118's concurrent protected PCC ledger conflict remains unresolved. Both histories and all claim/evidence bytes are preserved. Auto-merge stays off pending a legitimate canonical-maintainer reconciliation contract and actual-head gates. This integration dependency is separate from the mathematical work.

Separate local draft: terminal gain relative to D

Status. This appendix describes a separate five-statement local Lean draft. It is *outside* the frozen 57-statement source, 59-declaration axiom audit and running fresh replay. Exact source/log hashes are in

`../evidence/cost-relative-elaboration-001.json`

Data claim PCC-20261114-9b0604. Local elaboration exits 0 without warnings and central printed foundations are standard only [pcc:unverified].

Finite scale inequality. Let the terminal level be $2k$, with $2k+1$ retained levels, all actual schedule inequalities nonnegative, and $2k \le a$. Assume the actual diagonal polynomial obeys

$$\deg W_{xx} \le (2L-1) + KL, \qquad L = 2^a.$$

The already frozen actual polynomial budget implies $F_{4k+1} m_{2k} \le KL$. Standard Fibonacci addition gives

$$F_{n+4} = 2F_n + 3F_{n+1} \ge 5F_n, \qquad F_{4k+1} \ge 5^k.$$

The exact power split and the actual budget therefore give

$$L = 4^k 2^{a-2k}, \qquad 5^k m_{2k} \le K 4^k 2^{a-2k}.$$

All denominators below are positive; rational division yields

$$\boxed{\frac{m_{2k}}{D_{2k}} \le K(4/5)^k}, \qquad D_{2k} = 2^{a-2k}.$$

The primary retained-node application also keeps its usual $2k+1 \le a$ and other source hypotheses [pcc:unverified].

Proof map. All five declarations are in namespace `OrientationBudgetScale`, file `../evidence/cost-draft-002.lean.txt`:

- `fibonacci_four_growth`;
- `fibonacci_geometric`;
- `actual_terminal_geometric`;
- `power_split`;
- `actual_terminal_relative_scale`.

This is a finite necessary bound on the terminal multiplicity under the specified schedule and actual degree cap. It bounds the multiplicity contribution, not the whole starting valuation w, which also contains $a + 1 - 2k$. It does not establish the actual nonzero-mode inventory, account for possible coefficient sparsity/correlations, or prove an asymptotic runtime or algorithm lower bound [pcc:unverified].

The failed first draft and raw compiler outputs remain archived. The repair normalizes product associativity and removes over-general tactics without changing hypotheses or suppressing a linter.